Module 3: Conditional Probability — Practice

Lesson 2 | Calculate, justify, and interpret

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Preparation

Read Lesson 2. Use its Restaurant events and supplier model inputs, retaining their stated business settings and assumptions. Refer to the English–Arabic introductions there; no additional observations are provided here.

1. Which reference group?

Using the Restaurant CSV, calculate $P(A \mid B)$, $P(B \mid A)$, and $P(A \cap B)$. Show the numerator and denominator for each. Interpret each in one sentence.

2. Independence and exclusion

1. Compare $P(A \cap B)$ with $P(A)P(B)$. Are the events independent for uniform selection from these records?
2. Explain why this is not an independence hypothesis test for a larger population.
3. Explain why Very Good and Excellent ratings are mutually exclusive but are not independent when both have positive probability.

3. Two selections

Find the probability that two records are both Very Good when sampling without replacement. Compare it with independent selection with replacement. State the changed second-selection denominator and numerator.

4. Supplier evidence

Use supplier shares 0.65 and 0.35 and bad-part rates 0.02 and 0.05 from the lesson.

1. Calculate each supplier-and-bad joint probability and the total probability of a bad part.
2. Calculate both supplier probabilities conditional on a bad part and check that they sum to 1.
3. Calculate the probability a part comes from supplier 2 **given that it is good**.
4. Explain why 0.05 is not the answer to the question “Which supplier produced this bad part?”
5. State what raw inspection information is unavailable.

Submit a table of results with reference groups and assumptions. Return to the lesson.